

Positive Expectations Generate $\mathcal{W}$

Run fa_banach_001, agent agent_lane_11

17 August 2026

Source Question and Status

Douglas and Nowak work over the real scalars and put $\mathcal{C} = \ell_\infty(G)$. They define $\mathcal{W}$ as a weak-* closed subspace of

$$\mathcal{L}(\mathcal{L}(\ell_u(G), \mathcal{C}), \mathcal{C})$$

and let $\mathcal{M}_+$ be the set of normalized expectations in $\mathcal{W}$ whose restriction to $\ell_\infty(G, \mathcal{C})$ is positive. Their Question 5.2 asks whether every $\Xi \in \mathcal{W}$ can be represented as

$$\Xi = cM - c'M', \quad M, M' \in \mathcal{M}_+, quad c, c' \geq 0$$

[1, p. 17].

INVARIANT EXPECTATIONS AND BOUNDED COHOMOLOGY

17

5.2. Relation between $\mathcal{M}$ and $\mathcal{W}$. It is also interesting to investigate the relation between the set of positive expectations $\mathcal{M}_+$ (in the sense of remark 3.9) and the module $\mathcal{W}$. One possibility is that $\mathcal{W}$ is generated by $\mathcal{M}_+$ in the sense that for every $\Xi \in \mathcal{W}$ there exist $M, M' \in \mathcal{M}_+$ and constants $c, c' \in [0, +\infty)$ such that

$$\Xi = cM - c'M'.$$

Question 5.2. Is $\mathcal{W}$ generated by $\mathcal{M}_+$ in the above sense?

REFERENCES

This packet gives an affirmative answer for every finitely generated group. The proof uses only the definitions of $\mathcal{W}$ and $\mathcal{M}_+$; exactness and group invariance are not needed. Status: *candidate full solution, likely valid*, subject to human review.

Definitions

An element $\xi \in \ell_u(G)$ is written as a family $\xi = (\xi_g)_{g \in G}$ with $\xi_g \in \mathcal{C}$, and

$$\|\xi\|_u = \left\| \sum_{g \in G} |\xi_g| \right\|_{\mathcal{C}}.$$

It induces the evaluation operator

$$\widehat{\xi}(T) = T(\xi), \quad T \in \mathcal{L}(\ell_u(G), \mathcal{C}),$$

and the source proves that $\xi \mapsto \widehat{\xi}$ is isometric. Let

$$\mathcal{W}_{00} = \left\{ \xi : \# \text{supp } \xi < \infty, \sum_g \xi_g = c1_G \text{ for some } c \in \mathbb{R} \right\}.$$

Then $\mathcal{W}$ is the weak-* closure of $\widehat{\mathcal{W}_{00}}$. An expectation is an $M \in \mathcal{W}$ with $M(\mathbf{1}_G) = 1_G$, and $M \in \mathcal{M}_+$ when, additionally, its restriction to $\ell_\infty(G, \mathcal{C})$ is positive. Here $\mathbf{1}_G$ denotes the constant coefficient family used in the source.

Proof Intuition

Pointwise Jordan decomposition nearly settles the question. For finite ξ , its positive and negative coefficient families a, b satisfy $\xi = a - b$, but their total masses $A = \sum a_g$ and $B = \sum b_g$ can vary over G and hence cannot be normalized by scalar coefficients. Since $A - B$ is scalar, the same positive slack $R1_G - A$ can be added to both families at one group coordinate. Their difference is unchanged, while their new masses become the scalars R and $R - c$. A uniform choice of R works along a weak-* approximating net, so compactness passes the finite decompositions to $\mathcal{W}$.

Main Result

Theorem 1. *For every $\Xi \in \mathcal{W}$ there exist $M, M' \in \mathcal{M}_+$ and $c, c' \in [0, \infty)$ such that*

$$\Xi = cM - c'M'.$$

Consequently, the answer to Question 5.2 of [1] is yes.

Proof. Choose a net $\xi_\alpha \in \mathcal{W}_{00}$ such that $\widehat{\xi}_\alpha \rightarrow \Xi$ weak-*. Write

$$\sum_g (\xi_\alpha)_g = c_\alpha 1_G. \tag{1}$$

A weak-* convergent net in a dual Banach space is norm bounded. Since the evaluation embedding is isometric, there is $K < \infty$ with

$$\|\xi_\alpha\|_u \leq K \quad \text{for every } \alpha. \tag{2}$$

Fix once and for all a scalar $R > K$.

Take pointwise positive and negative parts in the real Banach lattice $\mathcal{C}$:

$$a_{\alpha,g} = ((\xi_\alpha)_g)^+, \quad b_{\alpha,g} = ((\xi_\alpha)_g)^-.$$

Set $A_\alpha = \sum_g a_{\alpha,g}$ and $B_\alpha = \sum_g b_{\alpha,g}$. Then

$$A_\alpha - B_\alpha = c_\alpha 1_G, \quad 0 \leq A_\alpha \leq A_\alpha + B_\alpha = \sum_g |(\xi_\alpha)_g| \leq K 1_G. \tag{3}$$

Thus $r_\alpha = R1_G - A_\alpha$ is positive. If e is the identity of G , define finite coefficient families

$$\begin{aligned} p_{\alpha,g} &= a_{\alpha,g} + \mathbf{1}_{\{e\}}(g)r_\alpha, \\ q_{\alpha,g} &= b_{\alpha,g} + \mathbf{1}_{\{e\}}(g)r_\alpha. \end{aligned}$$

They are positive and their difference is ξ_α . Moreover, using (3),

$$\sum_g p_{\alpha,g} = R1_G, \quad \sum_g q_{\alpha,g} = (R - c_\alpha)1_G. \tag{4}$$

The inequality $|c_\alpha| \leq \|\xi_\alpha\|_u \leq K < R$ shows that both normalizing scalars are positive. Therefore

$$M_\alpha = R^{-1}\widehat{p}_\alpha, \quad M'_\alpha = (R - c_\alpha)^{-1}\widehat{q}_\alpha$$

belong to $\mathcal{M}_+$. Indeed, they lie in $\widehat{\mathcal{W}}_{00}$, are normalized by (4), and are positive on $\ell_\infty(G, \mathcal{C})$ because all their coefficients are positive. Their norms are one: for a positive coefficient family the ℓ_u -norm is its total scalar mass. We have the exact finite decomposition

$$\widehat{\xi}_\alpha = RM_\alpha - (R - c_\alpha)M'_\alpha. \quad (5)$$

The pair (M_α, M'_α) lies in the product of two weak-* compact unit balls. Pass to a simultaneous subnet converging weak-* to (M, M') . Since $\mathcal{W}$ is weak-* closed, $M, M' \in \mathcal{W}$; evaluation at $\mathbf{1}_G$ shows that both remain normalized. Positivity also passes to the limit: the positive cone of $\mathcal{C} = \ell_\infty(G)$ is weak-* closed, so $M(f), M'(f) \geq 0$ for every positive $f \in \ell_\infty(G, \mathcal{C})$. Hence $M, M' \in \mathcal{M}_+$.

Finally, (1) and weak-* convergence show that $c_\alpha \rightarrow c$ for a scalar c and $\Xi(\mathbf{1}_G) = c\mathbf{1}_G$. Taking limits in (5) gives

$$\Xi = RM - (R - c)M'.$$

Since $|c| \leq K < R$, both coefficients are nonnegative. This is the required representation. $\square$

Scope, Novelty Check, and Verification

The theorem proves algebraic generation, which is stronger than merely placing $\mathcal{W}$ in the weak-* closed linear span of $\mathcal{M}_+$. It does not answer the source's separate, open-ended question about uniqueness of invariant expectations. The notation matters: immediately before Question 5.2 the source defines $\mathcal{M}$ using only conditions (a) and (b) of invariant expectations, so members of $\mathcal{M}_+$ need not themselves be group invariant.

The cheap run indexes contained no answer. A bounded search through 17 August 2026 used the exact question, the title and authors, and the phrases *positive expectations generate $\mathcal{W}$* and *invariant expectations exact groups*. It found the source, its published version, and later cohomological applications, but no answer to Question 5.2.

The companion script performs 500 randomized finite-dimensional checks of the common-slack identities. These are consistency checks only; the weak-* argument above is the proof.

Human Review Recommendation

Verify the isometric embedding used in (2), the simultaneous weak-* subnet, and weak-* closedness of positivity on the restricted module. The finite common-slack decomposition itself is pointwise algebra.

References

- [1] R. G. Douglas and P. W. Nowak, *Invariant expectations and vanishing of bounded cohomology for exact groups*, Journal of Topology and Analysis **3** (2011), 89–107; arXiv:1001.0718.